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“Air That Once Was Breath” Part 1: Wildfire-Smoke-Induced Mechanisms of Airway Inflammation – “Climate Change, Allergy and Immunology” Special IAAI Article Collection: Collegium Internationale Allergologicum Update 2023

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Abstract

Background: Wildfires are a global concern due to their wide-ranging environmental, economic, and public health impacts. Climate change contributes to an increase in the frequency and intensity of wildfires making smoke exposure a more significant and recurring health concern for individuals with airway diseases. Some of the most prominent effects of wildfire smoke exposure are asthma exacerbations and allergic airway sensitization. Likely due to the delayed recognition of its health impacts in comparison with cigarette smoke and industrial or traffic-related air pollution, research on the composition, the mechanisms of toxicity, and the cellular/molecular pathways involved is poor or non-existent.

Summary: This review discusses potential underlying pathological mechanisms of wildfire-smoke-related allergic airway disease and asthma. We focused on major gaps in understanding the role of wildfire smoke composition in the development of airway disease and the known and potential mechanisms involving cellular and molecular players of oxidative injury at the epithelial barrier in airway inflammation. We examine how PM_{2.5}, VOCs, O₃, endotoxin, microbes, and toxic gases may affect oxidative stress and inflammation in the respiratory mucosal barrier. We

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Author Contributions

W.S.B. drafted the first draft of the manuscript; R.J.S., G.K.S., G.R.T. III, H.J. A.A.Z., and A.H. wrote various sections and contributed to others. A.H. arranged the final version and finalized the paper. All authors participated in editing and approved the final draft.

Conflict of Interest Statement

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discuss the role of AhR in mediating smoke's effects in alarmin release and IL-17A production and how glucocorticoid responsiveness may be impaired by IL-17A-induced signaling and epigenetic changes leading to steroid-resistant severe airway inflammation.

Key Message: Effective mitigation of wildfire-smoke-related respiratory health effects would require comprehensive research efforts aimed at a better understanding of the immune regulatory effects of wildfire smoke in respiratory health and disease.

Keywords

Wildfire smoke; Oxidative stress; Epigenetics; Aryl hydrocarbon receptor; IL-17A

Introduction

Among its recognized health effects, wildfire-smoke-induced asthma exacerbations and allergic airway sensitization count prominently [1-6]. Components of wildfire smoke carry specific pathogens and allergens [7,8] and act as powerful toxic irritants. How wildfire smoke components may damage the respiratory epithelial barrier and sensitize for allergic and inflammatory response is poorly understood. While the immediate consequences (fatalities, emergency room visits, and stress caused by the destruction of homes, livelihoods, and mass evacuations) are readily acknowledged, little attention has been paid to the medium- and long-term respiratory health effects of wildfires. Of great additional concern is the question of how vulnerable populations, children, the elderly, pregnant women, and those from marginalized communities (due to ethnic or socioeconomic status), become at high risk of developing allergic airway disease upon repeated wildfire exposures [8-10]. Here, we aimed to discuss currently available information with a focus on potential cellular and molecular pathways affecting respiratory mucosal barrier function and airway inflammation.

Wildfires generate plumes of air pollutants from smoke that quickly spread reducing air quality to hazardous levels across vast areas [11]. The source and composition of inhaled smoke are important as it may elicit differential respiratory effects [4, 9, 12-30]. However, it is not well understood how specific smoke components may induce differential disease outcomes, what are the most toxic components, and how those work. For this reason, workshop reports from the American Thoracic Society in 2021 [31] and National Academies of Sciences in 2022 [32] identified health effect studies on wildfire smoke fuel type, combustion phase (i.e., flaming vs. smoldering), concentration, and exposure duration (i.e., acute vs. repeated episodic) as top priority research areas. We reference here a summary list that was prepared and updated in 2020 by the Office of Environmental Health Hazard Assessment (OEHHA) on inhalational toxicants for their acute, 8-h, and chronic reference exposure level (<https://oehha.ca.gov/air/general-info/oehha-acute-8-hour-and-chronic-reference-exposure-level-rel-summary>) [33]. The most studied major wildfire smoke components with relevance in eliciting and exacerbating allergic and chronic airway diseases are particulate matter (PM), volatile organic compounds (VOCs), and ozone (O₃). Over the last 3 years in Northern California, the highest ever concurrent O₃ and PM concentrations were recorded, often for weeks at a time.

For review purposes, we defined wildfires as wild and prescribed forest fires, tropical deforestation fires, peat fires, agricultural burning, and grass fires. We compared some findings on wildfire smoke, combustion of indoor household biomass fuels for cooking and heating, and ambient air pollution. We also used information from controlled in vivo and in vitro experimental exposure studies on the effects of PM, VOCs, and O₃ from varying sources. Our search covered the period between 1980 and 2023. We used “PubMed,” “Web of Science,” and Google search engines, and we referenced peer-reviewed original publications, reviews, government publications, and news articles (the latter only to highlight very recent wildfire occurrences).

Particulate Matter

Wildfire smoke contains particles of varying size, but the ones with pathological significance should be smaller than 10 µm in diameter (PM₁₀; coarse particles) as those could pass through the upper respiratory tract and enter the proximal and distal airways and alveoli. Most (>90%) of all particle mass emitted from wildfires are smaller than 2.5 µm in diameter (PM_{2.5}) and are categorized as “fine” or “ultrafine,” if smaller than 0.1 µm in diameter (PM_{0.1}). A review of 45 epidemiological studies across the globe (1986–2015) found that levels of PM₁₀ were 1.2–10 times higher due to wildfire smoke compared to non-fire periods and/or locations. In over 90% of these studies, wildfire smoke was significantly associated with a risk of respiratory morbidity [13]. Studies on the onset of asthma symptoms, oral steroid medication, mean daily symptom count, and mean daily dose of reliever medication showed similar associations with inhaled PM₁₀ and PM_{2.5} [34] during wildfire events suggesting that coarse and fine PM are similarly responsible for eliciting acute asthma symptoms.

Wildfire PM was shown to cause cell damage and cell death via oxidative stress in mouse macrophages [35, 36]. The coarse PM₁₀ fraction in these studies was more proinflammatory and had several-fold higher concentrations of the oxidation products of phenanthrene and anthracene, phenanthraquinone and anthraquinone, than the fine PM fraction on an equal-dose basis [35]. These data suggested a significant role for atmospheric photochemistry in the formation of secondary pollutants and the possibility of significant additional sources of toxicity depending on the chemical composition of PM. The mechanisms through which oxidative stress, polycyclic aromatic hydrocarbon (PAH) signaling, and proinflammatory cytokines affect the pathogenesis of asthma are discussed below.

The role of endotoxin in wildfire particulate toxicity has also been examined given its presence in wildfire smoke and its ability to cause inflammation via toll-like receptor pathways. To assess the endotoxin’s role in inflammation caused by wildfire particulate, investigators compared the effects of wildfire smoke extract on human [37] and mouse [38] bronchial epithelial cells, with or without polymyxin B pretreatment (used to bind and inactivate endotoxin). Their results suggest that endotoxin played only a minor role in wildfire-mediated macrophage toxicity. Other investigators using similar methods but with endotoxin from North Carolina peat wildfire particulates found that it had a greater portion of toxicity in female CD-1 mice [39]. Dunn et al. [40] demonstrated that mice exposed to smoldering plywood smoke had CXCL-dependent neutrophil influx that was associated with

impaired bacterial clearance and greater susceptibility to *Pseudomonas* pneumonia. These results suggest that wildfire smoke containing endotoxin may modulate immune function in the airways. However, more research is needed to clarify the mechanisms and significance of this effect.

Wildfire smoke has been linked with exposure to plant-derived irritants, pollen and fungal particles potentially evoking immune reaction and allergic sensitization. For example, smoke from fires that burn through poison oak, poison ivy, and poison sumac may contain traces of the irritant urushiol that induces a delayed-type hypersensitivity response upon inhalation. Such exposure can cause severe respiratory distress [41]. Whether wildfire smoke containing allergenic particles could directly induce respiratory allergies is not clear [2]. It has been however reported in a longitudinal observational cohort study design using meteorological data and a patient-level clinical dataset from a local outpatient allergy clinic ($n = 842$) that there were decreases in respiratory peak flow among allergy clinic patients 1 year after each wildfire event. This study raised the possibility that wildfire smoke itself may elicit a delayed immune response or that it enhances allergic sensitization to other allergens [4].

Kobziar and colleagues [42] used unmanned aircraft systems to analyze the aerosols above high-intensity forest fires in the western USA. They found aerosolization of viable microbes via wildfire smoke and identified both pathogenic and non-pathogenic fungal species using DNA analysis [42]. They also established that wildfire smoke contained about four-fold higher concentrations of microbes with 78% viability, five-fold higher taxon richness, and about three-fold enrichment of ice-nucleating particle concentrations in smoke, implying that wildfire smoke is an important source of diverse bacteria and fungi as well as meteorologically relevant aerosols [43]. These findings are important as they raise the possibility that wildfire smoke inhalation may affect the host's respiratory microbiome, and local immune milieu, potentially leading to the onset or exacerbation of airway disease.

It is important to understand that as the size of PM decreases, it can travel deeper into the bronchoalveolar spaces. PM_{2.5} can cross into the circulation, enter the brain and other organs, and cause direct long-term systemic or organ-specific damage. In a recent review by Kunovac et al. [44], the systemic molecular effects of fine and ultrafine particles from a wide range of sources (including wildfire smoke) have been summarized and highlighted. Inhalational exposures to PM_{2.5} and PM_{0.1} affect virtually all aspects of cellular functions systemically and in multiple different organs. Human and experimental animal studies showed that PM_{2.5} and PM_{0.1} elicited immune, inflammatory, oxidative, and endoplasmic reticulum stress and affected carcinogenic [45], metabolic, and epigenetic pathways [44].

Observational evidence from Southern California suggests that wildfire-specific PM_{2.5} is up to 10 times more harmful to human health than PM_{2.5} from other sources [24]. In comparing inflammatory responses between urban and wildfire-derived PM_{2.5} under in vitro experimental conditions, Nakayama and colleagues at UC Davis found that wildfire PM activated CYP1A1 (cytochrome P450, family 1, subfamily A, polypeptide 1) and CYP1B1 genes to a greater degree than urban (non-combustion-derived) PM [37]. The cytochrome P450 superfamily of enzymes mediates environmental stressors and is involved in numerous biological and pathological processes [46]. Nakayama Wong and colleagues

[37] proposed that the differences seen in the effects of wildfire smoke and urban PM were due to differing composition of PAHs that induce oxidative stress via redox cycling. Wildfire particulates also induced the oxidative stress-related gene heme oxygenase 1 (HMOX1, an antioxidant enzyme) that was markedly attenuated after the pretreatment of smoke extracts with deferoxamine, suggesting an important role for iron (Fe^{++}) in the generation of reactive oxygen species. By contrast, urban (non-wildfire) PM induced greater levels of the proinflammatory cytokines IL-1 α , IL- β , IL-8, CCL20, and GM-CSF than wildfire smoke PM. The authors concluded that wildfire PM as compared to urban PM causes cellular oxidative damage via a transition metal-catalyzed Fenton-like reaction (between Fe^{++} and hydrogen peroxide [H_2O_2]) to produce hydroxyl radicals as well as PAH-mediated redox cycling. Interestingly, other investigators found that wildfire PM may have a PAH composition more prone to cause oxidative stress and chemical components that are more toxic to the lung than equal doses of PM collected from ambient air from the same region [47].

VOCs and Toxic Gases

Different fire substrates produce disparate effects of combustion by-products in wildfire smoke and affect the ability to predict wildfire smoke toxicity and its linkage with allergy and asthma. Urban fires or wildfires in the wildland-urban interface are particularly challenging for these reasons. Such fires in addition to coarse and fine PM also emit large amounts of toxic gases such as SO_2 , nitric oxide (NO), NO_2 , CO, CO_2 , as well as metals, PAHs, and other VOC (e.g., aldehydes, n-alkanes) [48, 49]. Among the toxic gases, SO_2 is released primarily from the combustion of sulfur-containing coal and oil and it is not considered a major component of wildfire smoke. However, we would mention here that people prone to allergies, especially allergic asthma, can be extremely sensitive to even small amounts of inhaled SO_2 as it is an irritant that penetrates deep into the air spaces where it is converted into bisulfite and interacts with sensory receptors, causing bronchoconstriction (reviewed in [50]).

VOC and toxic agents are released by fire from building construction materials, interior furnishings including plastics, solvents, glues, metals, formaldehyde, and halogens during high-temperature combustion [49]. These components (also referred to as Hazardous Air Pollutants or Toxic Air Contaminants by the California Environmental Protection Agency [CalEPA]) may directly act in concert as respiratory irritants that exacerbate asthma [33, 51, 52]. VOC is particularly hazardous for infants, children, pregnant women and their fetuses, elderly persons, those with existing lung, heart, or liver diseases, and persons engaging in physical activity, such as firefighters. Acetaldehyde, acrolein, formaldehyde, and benzene are of specific interest because of their differential impact on infants and children compared to adults. These contaminants have a cumulative toxic effect and can be present in wildfire smoke in concentrations above the limit determined by the Office of Environmental Health Hazard Assessment (OEHHA) as toxic [33]. Recent systematic reviews on the specific effects of VOC exposure (that were mostly studied in the context of indoor air pollution) uncovered strong associations with asthma [53, 54]. Metanalysis of 32 studies provided robust evidence for the role of VOC by pooled risk ratios for asthma of 1.08 (CI: 1.02–1.14), 1.02 (1.00–1.04), and 1.04 (1.02–1.06) per 1 $\mu\text{g}/\text{m}^3$ increase of benzene, toluene, and

p-dichlorobenzene, respectively [53, 54]. There was also a pooled risk ratio of 1.12 (1.05–1.19) per 1 $\mu\text{g}/\text{m}^3$ increase in benzene exposure of pregnant women for low newborn birth weight. Additionally, a metaanalysis of 13 papers demonstrated a pooled odds ratio between formaldehyde exposure and asthma. Each 10 $\mu\text{g}/\text{m}^3$ increase in formaldehyde exposure was significantly associated with a 10% increase in the risk of asthma in children (OR =1.10 [CI: 1.00–1.21]) [53]. Formaldehyde exposure was also associated with an increased risk of asthma in adults (1.81 [1.18–2.78]) when separating the high-exposure adult group (FA >22.5 $\mu\text{g}/\text{m}^3$) from the rest of the population [54].

VOCs together with nitrogen oxides may also indirectly contribute to the harmful effects of wildfire smoke since through reactions with sunlight, these molecules catalyze the generation of ground-level (tropospheric) O_3 . NO and NO^2 are reactive atmospheric pollutants associated with wildfire smoke and known for their oxidant properties directly or in the form of the more noxious peroxyxynitrite (ONOO^-). Exposure to NO_2 by itself was shown to promote allergic airway inflammation and increase asthma susceptibility according to a study by Lu et al. [55]. Using a model of ovalbumin-induced allergic airway inflammation, the authors found that NO^2 exposure enhanced inflammatory changes, airway hyperreactivity, and remodeling. These effects were associated with increased reactive oxygen species and malonaldehyde, and decreased glutathione expression, indicating high levels of oxidative stress. De Sousa and colleagues [56] investigated fire-pollutant meteorological variables and their impact on cardio-respiratory mortality in Portugal during wildfire season. The authors studied $\text{PM}_{2.5}$ and PM_{10} , CO , NO_2 , O_3 , temperature, relative humidity, wind speed, aerosol optical depth, and mortality rates and assessed pollutant-burning interaction and atmospheric-pollutant interaction. Using cluster analysis, they showed that colder and wetter months and higher NO_2 concentration correlated with respiratory symptoms and pneumonia, while warmer and drier months and higher PM_{10} , $\text{PM}_{2.5}$, CO , and O_3 concentrations correlated with respiratory and cardiovascular symptoms and airway disease.

CO and CO_2 are products of incomplete combustion of biomass and fossil fuels. Evidence suggests an association between exposure to CO and moderate or severe asthma exacerbations in adults. In a study conducted in China, a total of 72,430 hospitalized cases showed significant positive exposure-response relationships between ambient CO exposure and respiratory disease [57]. For each 1 mg/m^3 increase in CO concentration (lag0-2), hospitalizations for total respiratory diseases, asthma, and chronic obstructive pulmonary disease (COPD) increased by 13.56% (95% CI: 6.76%, 20.79%), 17.74% (95% CI: 1.34%, 36.8%), and 12.45% (95% CI: 2.91%, 22.87%), respectively. Women were more susceptible to ambient CO exposure-associated hospitalizations for asthma and lower respiratory tract infection.

Pertinent to allergic airway sensitization, CO - and CO_2 - level increases in the atmosphere were also linked to prolonged pollen seasons, larger quantity, and allergenicity of pollen produced by plants (reviewed in [58]). Birch pollen extracts from trees grown in warmer temperatures had stronger IgE binding intensity, while ragweed pollen production increased with higher CO_2 levels. *Alternaria* species growth and allergen production were also increased in an enhanced CO_2 environment [58].

Ozone

The importance of O₃ in airway disease is supported by the fact that in cities with high O₃ levels, people have an over 30% increase in risk of dying from airway disease [59] and children playing outdoor sports have a three times greater chance of developing asthma [8, 60]. O₃, a direct source of oxidative stress (reviewed by Enweasor et al. [61]), can reduce lung function and induce inflammation of the airways, chest pain, coughing, wheezing, and shortness of breath, even in healthy people. Patients with existing airway disease are highly susceptible to the respiratory effects attributed to O₃ exposure that can lead to increased use of medication, school absences, hospital admissions, and emergency room visits for asthma and COPD.

Experimental O₃ exposure was shown to induce significant airway inflammation and hyperreactivity in mouse models of asthma [62-75], asthmatic rhesus macaques [76-79], healthy human subjects [80-83], and patients with asthma and COPD [20, 84-90]. Urban O₃ concentrations steadily declined in the Eastern USA in recent decades because of effective regulations limiting the release of its chemical precursors [91]. O₃-attributable mortality worldwide, however, increased by 46% from 2000 to 2019 [92]. Climate-change-related wildfire activities with heightened VOC and nitrogen oxide emissions may significantly contribute to this trend. PM concentrations in the air are strongly related to O₃ concentrations during wildfires.

The Epithelial Barrier Hypothesis

In a recent targeted proteomic analysis, Aguilera et al. [93] revealed significant changes in molecules critical for immune and barrier function in a study conducted during a 10–12 day-long continuous wildfire smoke exposure period in the San Francisco Bay area. The authors found an association of the zonulin family peptide (zonulin, a molecule responsible for regulating paracellular epithelial barrier tight junctions) with T helper 2 cells, major drivers of allergic sensitization. Thus, wildfire smoke may induce epithelial barrier injury leading to tolerance breakdown [94], increased susceptibility to allergic airway sensitization, and development of asthma and COPD [28] (Fig. 1).

In an ex-vivo airway epithelial model [95], wildfire smoke extract increased cellular permeability and blunted autophagy to a greater extent than cigarette smoke. These data suggest that cell fate and epithelial barrier integrity are significantly affected by wildfire smoke. In vitro wildfire smoke exposure also caused a direct and rapid cell death of macrophages [38, 96], a major cell type at the airway epithelial barrier. The level of macrophage toxicity correlated with oxidative stress and NF-κB signaling. These effects were likely due to heat-labile organic compounds and not endotoxin content [38]. Oxidative stress in the airways during wildfire smoke inhalation can be induced *exogenously* by inhalation of PM and gaseous components, specifically O₃, and *endogenously*, from reactive oxygen and nitrogen species released from activated inflammatory cells, particularly neutrophils, macrophages, and eosinophils (Fig. 1).

Role of Oxidative Stress

Studies investigating the effects of wildfire smoke components including PM and O₃ often show that the induction of asthmatic airway changes is associated with oxidative stress. When pro- and antioxidant mechanisms are in an imbalance, the resulting oxidative stress causes lipid peroxidation of unsaturated phospholipids, glycolipids, and cholesterol, affecting the function of the pulmonary surfactant phospholipids [97] and cell membranes [98, 99], methionine, and sulfhydryl group oxidation in proteins, depletion of antioxidants, and DNA damage [100-102]. Oxidative stress is characterized by the presence of *free radical species* such as oxyl radicals, peroxy radicals, and hydroxyl radicals derived from Fe⁺⁺-mediated reduction of H₂O₂ or *non-radical species* such as singlet oxygen, O₃, and ONOO⁻ produced by the reaction between NO and superoxide radicals [100-102]. While NO and O₃ cause direct oxidative damage upon inhalation, other reactive chemical species, such as H₂O₂, superoxide anion, hydroxyl and hypohalite radical, are formed endogenously by inflammatory cells through enzymatic action [102].

Indeed, a variety of cells that also participate in the allergic inflammatory response of asthmatic airways, e.g., macrophages, mast cells, neutrophils, eosinophils, and basophils (Fig. 2.), generate superoxide anion, which is rapidly converted to H₂O₂ by the enzyme superoxide dismutase, or hypohalite radical in the presence of Fe⁺⁺ as a secondary reaction [102-104]. Activated neutrophils and macrophages produce myeloperoxidase, while eosinophils specifically release eosinophil peroxidase. While both of these enzymes are members of the mammalian peroxidase family and use halide anions (I⁻, Br⁻, Cl⁻) for activation, myeloperoxidase mainly uses chloride and results in the formation of chlorotyrosine, whereas eosinophil peroxidase selectively uses bromide yielding bromotyrosine as a product. Because of this selectivity, high levels of urine bromotyrosine serve as a biomarker of eosinophil activation in asthma [105].

Like O₃, NO is a highly reactive atmospheric pollutant associated with wildfire smoke, with oxidant properties directly or in the form of the more noxious ONOO⁻. NO, however, can also be produced endogenously, particularly in the lung, through the enzymatic action of NO synthase. NO functions as a vasodilator, bronchodilator, neuro-transmitter, and inflammatory mediator (reviewed in [106]). Patients with asthma have high levels of NO in their exhaled breath and high levels of inducible NO synthase enzyme expression in the epithelial cells of their airways. Exhaled NO therefore is also an indirect marker for upregulation of airway inflammation and is used to diagnose eosinophilic, type 2 asthma.

Inflammatory Mechanisms

Oxidative stress in the airways leads to the release of alarmins including cytokines (IL-1 β , IL-6, IL-23, IL-25, IL-33, TNF- α , and TSLP) and other inflammatory mediators such as arachidonic acid derivatives and damage-associated molecular patterns triggering a cascade of proinflammatory changes in structural and immune cells in the respiratory mucosal tissue [61, 68, 71, 78, 107-114]. Such changes are also hallmarks of severe asthma exacerbations [111] (Fig. 2.).

In the healthy lung, the primary inflammatory cells recruited to the airways following oxidative injury (studied in response to O₃ [81, 82] or endotoxin-containing PM [39]) are the neutrophils. These cells appear in the airways within minutes and accumulate in significant numbers as early as 1–2 h after exposure [64, 68, 76]. In wildfire smoke-exposed infant monkeys [115] and O₃-exposed healthy human subjects [81, 83, 89], such inflammatory changes were associated with a decrease in lung function. Importantly, when allergen-sensitized mice, asthmatic rhesus macaques, or allergic human subjects were exposed to O₃, a marked influx of both eosinophilic and neutrophilic granulocytes was observed [68, 78, 116, 117] with greatly amplified lung function impairment, indicating the pathological significance of O₃ exposure in asthma.

Among the many mediators released in response to oxidative damage of the airway epithelium IL-25, IL-33 and TSLP play an important role in priming type 2 airway inflammation and causing severe asthma exacerbation [68, 72, 108–113] (Fig. 2). Specifically, our laboratory and others previously found that IL-33 transcription as well as release was upregulated by oxidative damage in the lung in a time-dependent manner [68, 110]. Interestingly, Michaudel and colleagues [110] also showed that the absence of IL-33 or neutralization of the IL-33 receptor (ST2) although reduced airway responsiveness led to enhanced epithelial cell injury with protein leak and increased neutrophilia that was reversed by replacement of IL-33. These results suggested that like many components of the oxidative stress/inflammation cascade, IL-33 may have both pro- and anti-inflammatory roles.

Destruction of Anti-Inflammatory and Antioxidant Mechanisms

Oxidative stress not only causes lipid peroxidation, methionine, and sulfhydryl group oxidation in proteins, and DNA damage, but also depletes antioxidants, thereby generating an oxidant imbalance [100, 101]. In addition to the nutritionally obtained ascorbic acid, α -tocopherol, lycopene, and β -carotene, the major endogenously produced antioxidants that fight against the effects of oxidative stress are the antioxidant enzymes such as superoxide dismutase, catalase, and glutathione. While there are no studies investigating the effects of wildfire smoke on antioxidant expression, Sahiner and colleagues [102] found that antioxidants were expressed at significantly lower levels in asthmatic children compared with healthy controls (reviewed in [102]).

Against inflammatory injury, the lung mounts immuno-protective mechanisms such as production of the epithelial-cell-derived lung collectins, surfactant protein A and D. We and others showed that SP-D plays an important anti-inflammatory function in allergic airway inflammation [71, 116, 118, 119]. Oxidative stress however was shown to destroy the biologically active quaternary and tertiary structure of this molecule [70, 120, 121]. Indeed, a de-oligomerized form of SP-D can readily enter the circulation from the lung during lung injury. We have recently found that wildfire smoke exposure of healthy volunteers caused a significant increase in the levels of circulating SP-D (unpublished information) indicating wildfire-smoke-induced oxidative lung injury and the biomarker potential of this molecule.

The mainstay of treatment in asthma and other chronic inflammatory diseases is formed by glucocorticoid therapy. However, studies on mice [69], dogs [122], rhesus macaques [123], and asthma patients [84, 124] showed diminished effectiveness of glucocorticoids in inhibiting exacerbation of asthmatic airway inflammation caused by exposure to air pollutants. Because oxidative stress causes neutrophil influx in asthma and because neutrophils are poorly responsive to glucocorticoids [125], wildfire-smoke-induced asthma exacerbations could potentially be resistant to glucocorticoid treatment, generating an important clinical problem (reviewed in [61]).

Role of the Aryl Hydrocarbon Receptor

The aryl hydrocarbon receptor (AhR) mediates the effects of wildfire smoke components while regulating asthma-related genes. AhR is expressed intracellularly in myeloid, lymphoid, and structural cells and functions as a transcription factor (TF) activated by environmental chemicals (such as aromatic hydrocarbons) or endogenous indole derivatives (such as kynurenine) [126]. AhR is critical for a wide range of immune functions including the maintenance of innate and adaptive cell populations at mucosal barrier sites, and the control of inflammation at steady-state or during ongoing inflammatory responses in asthma is involved in cell differentiation, cell adhesion, mucus and cytokine production [72, 127, 128]. Upon ligand binding, the AhR complex translocates into the nucleus and heterodimerizes with AhR nuclear translocator to induce gene transcription. AhR is a main activator of the genes encoding cytochrome P450 and the cytokines IL-17A and IL-22 (Fig. 3). The effects of AhR on immune cell differentiation (including Th17 or Treg polarization) are influenced by the nature of the ligand and the local cytokine milieu, and it could be either protective or pro-asthmatic (reviewed in detail by Poulain-Godefroy et al. [126]).

AhR interferes with the action of nuclear factor-kappa B (NF- κ B), a proinflammatory TF in different ways. For example, NF- κ B induces AhR expression, but AhR then regulates NF- κ B signaling [128]. By interacting with the function of other TFs, AhR promotes IL-22 (ROR γ t), IL-10, and IL-21 (cMaf) as well as Aiolos and its own expression (STAT3). Through Aiolos, AhR inhibits expression of the T-cell growth factor, IL-2 [128]. Thus, on the one hand AhR promotes Th17 cell differentiation; on the other, it induces Th17 cell plasticity into IL-10 producing protective Tr1 cells. While both IL-17A and IL-22 can elicit airway neutrophilia, IL-22 can also play a protective role when produced during epithelial or tissue damage. Chronic oxidative lung injury was reported to be associated with increased tryptophan and lipoxin A4 (activators of AhR), and recruitment of IL-17A and IL-22-expressing cells. T-cell-specific AhR depletion enhanced lung inflammation, indicating that oxidative stress activates AhR, which controls airway inflammation by reduction of IL-22 expression [72].

IL-17A and Glucocorticoid Responsiveness

IL-17A was implicated in wildfire-smoke-induced airway inflammation as one of the main players [129, 130]. The IL-17A and IL-22 genes are activated by the ROR γ t proinflammatory signaling pathway [131, 132]. Combustion-derived PM containing environmentally persistent free radicals induced IL-17-related airway inflammation as well

as activation of Cyp1a1 and Cyp1b1 gene expression in airway epithelial cells indicating the involvement of AhR. Using single-cell RNA sequencing analysis, combustion-derived PM exposure of the mouse lung showed that epithelial cells acquire a transcriptomic profile indicative of increased IL-17 signaling, AhR activation, EGFR signaling, and T-cell receptor and costimulatory signaling pathways. AhR activation was brought on by Ahr/AhR nuclear translocator and activation of tyrosine kinase c-Src, EGFR, and subsequently Erk1/2 pathways [130]. Thus, PM initiates mixed neutrophilic and eosinophilic asthma through epithelial, dendritic, and T-cell AhR activation involving IL-17A release (Fig. 3.).

While Th17 cells are identified as the main producers of IL-17A, in our in vivo studies, O₃-induced asthma exacerbation in mice did not show T-cell activation or migration of T cells into the lung prior to the O₃-prompted neutrophil influx [68]. These results implied that Th17 cells do not participate in IL-17A release in the early phases of the O₃ response. Mathews et al. [132] proposed that the source of IL-17A in response to acute O₃ exposure is the $\gamma\delta$ T cell. In addition, innate lymphoid cells were shown to be essential and sufficient to elicit the development of O₃-induced neutrophilia [68] and the ensuing airway hyperresponsiveness in mice. These studies suggest the importance of innate immune players in O₃-induced IL-17A pathways. Taken together, in addition to Th cells, both IL-17A and IL-22 can be produced by ILC3, $\gamma\delta$ T, and NK cells, after stimulation with IL-1 β , TGF- β , IL-6 or IL-23, and the TF ROR γ t [72]. These mechanisms are important because IL-17A-mediated neutrophilia in response to oxidative stress feeds back to a vicious cycle by releasing additional free radicals into the airways. The clinical relevance of these experimental findings is supported by the fact that increased IL-17A expression in the lung was associated with steroid-insensitive neutrophilic asthma in patients [133]. While glucocorticoid responsiveness varies among individuals depending on genetic, environmental, and immune factors, wildfire-smoke-induced *oxidative stress*, *inflammatory mediators*, and *epigenetic changes* [61, 104, 134, 135] can all reduce immune cell glucocorticoid responsiveness leading to persistent inflammation. IL-17 is particularly interesting in this mechanism for several reasons [136].

IL-17A through IL-17RA [137-140] can directly activate signaling pathways such as the NF- κ B p38-MAPK and PI3K pathway that antagonize GR function [134, 141]. IL-17A-induced NF- κ B activity was shown to consume and compete for transcription coactivators, essential for both transactivation of anti-inflammatory genes and transrepression of proinflammatory ones by the GR. NF- κ B can also interfere with GR nuclear translocation and DNA access (at glucocorticoid response elements) and can promote post-translational modifications (such as phosphorylation) and degradation of the GR. IL-17A was also shown to activate a dominant negative isoform (GR β) leading to reduced anti-inflammatory effects. Th17 cell differentiation was shown to be in fact upregulated by dexamethasone treatment in vitro and in mice in vivo [142], along with other factors like increased Bcl-2 expression, heightened ROR γ t activity, amplified Stat3 phosphorylation, and unique metabolic traits such as higher level of glucose and glutaminolysis [143]. Low levels of glucocorticoid-induced leucine zipper protein and histone deacetylase 2, and elevated IL-6, IL-23, IL-22, and STAT3 activation were described as main players in glucocorticoid resistance. Taken together, IL-17A potentially contributes in major ways to glucocorticoid resistance in wildfire-smoke-exposed populations especially those suffering from chronic

severe inflammatory airway disease, involving intricate molecular pathways and various immune and inflammatory cell types.

Epigenetic Modifications of Asthma-Related Gene Expression

Accumulating evidence obtained from studies on exposure to wildfire smoke, air pollution, and other toxic sources suggests that inflammation and oxidative stress are linked with epigenetic changes (including DNA methylation, histone modification, and microRNAs) [144-149]. Airborne PM has been shown to induce redox-sensitive signaling mechanisms and irreversible epigenetic changes that may persist through generations [150].

The role of extracellular vesicles in wildfire smoke response in the airways has recently been investigated by Carberry et al. [149]. Using in vitro and animal models of exposures to smoke condensate from peat and red oak, the authors found increases in miR-150, miR-183, miR-223-3p, miR-30b, and miR-378a in extracellular vesicles. These miRNAs were identified to regulate hypoxia and cell stress-related gene transcription. Wildfire smoke might contribute to the transmission of epimutations from gametes to zygotes by involving mitochondrial DNA, parental allele imprinting, histone withholding, and non-coding RNAs. Larger prospective studies using innovative, integrated epigenome-wide strategy are highly warranted to further investigate epigenetic inheritance and the associated effects of wildfire smoke on allergic airway disease [150].

DNA methylation was also associated with immune and inflammatory changes aggravating models of allergic airway inflammation [151-153]. The dynamic DNA methylation landscape can be rapidly altered (within hours) in peripheral leukocytes following PM_{2.5} exposure, linking systemic inflammation and oxidative stress [154, 155]. Indeed, DNA methylation changes were identified in inflammatory and oxidative stress response genes [156]. In rats, traffic-related PM exposure induced DNA methylation alterations in both blood and lung tissues that were dependent on exposure dose and duration, as well as the sex of the rat [157]. Most evidence in humans is based on experiments utilizing a heterogeneous mixture of leukocytes [154, 155, 158, 159], but specific loss of methylation in inflammatory genes and subsequent inflammatory responses, specifically in circulating T helper cells, was found in vivo after environmental challenge [160, 161]. A human intervention study of 2-h exposure to median 234.0 µg/m³ PM_{2.5} substantially modified DNA methylation in CD4⁺ T helper cells, with changes in genes involved in mitochondrial oxidative energy metabolism [162] suggesting these epigenetic changes as potential mediators of adverse health outcomes [163].

PM_{2.5} from different sources also induces significant DNA methylation changes in cultured bronchial epithelial cells at different doses and time points [164, 165]. Current and early life exposure to PM_{2.5} from outdoor and indoor environments has been associated with DNA methylation changes in nasal epithelial cells [164, 166, 167]. Brown and colleagues at UC Davis studied long-term baseline epigenetic changes associated with early-life exposure to wildfire smoke in rhesus macaques. Using whole genome bisulfite sequencing of nasal epithelial cells, they identified 3,370 differentially methylated regions annotated to genes significantly enriched for synaptogenesis signaling, protein kinase A signaling, and a

variety of immune processes. Some DMRs significantly correlated with gene expression differences, suggesting that early-life exposure to wildfire smoke leads to long-term changes in the methylome over genes impacting both the nervous and immune systems [148].

PM-sensitive CpG sites were also mapped to disease-related genes, including pulmonary disease, in peripheral blood leukocytes of 8,397 healthy subjects [168]. PM exposure was further linked with methylation in the CFTR cystic fibrosis gene. Long-term PM exposure was associated with altered DNA methylation in individuals with COPD [169]. Other studies similarly revealed abnormal DNA methylation in patients with asthma [166, 170-173] and COPD [174, 175]. Importantly, DNA methylation changes linked to wildfire PM_{2.5} in pregnant women and their relevance to respiratory function remain unstudied.

Two enzyme systems, namely, DNA methyltransferase and ten-eleven translocation (TET) proteins, methylate and demethylate DNA from C to 5-methylcytosine and back to C to establish, genome-wide, a methylation pattern required for gene regulation and cell fate decisions [176] (Fig. 4a). The potential role of gene-specific epigenetic pathways in mediating the effects of wildfire smoke is highlighted by the *Tet1* gene [166, 177]. Methylation of this gene promoter itself has been associated with childhood asthma and traffic-related air pollution (reviewed by Poulain-Godefroy et al. [126]).

TET family proteins may work as transcriptional activator or repressor through their enzymatic and non-enzymatic activity in multiple cellular processes. Our recent study on house dust mite-challenged *Tet1*^{-/-} mice (compared to *Tet1*^{+/+} mice) showed an increased allergic airway inflammation associated with a decrease in AhR-dependent protective *Cyp1a1* and *Aldh1a1* gene expression and an increase in interferon signaling. In human bronchial epithelial cells, activation of TET1 differentially regulated interferon and AhR signaling pathways, with predicted binding of transcriptional factors with relevant functions in their promoters, and the presence of histone marks generated by histone enzymes. TET1 forms a complex with hMOF (a histone acetyltransferase) and SIN3A (a scaffold protein), promotes acetylation of H4K16ac, and regulates genes in DNA repair. Thus, TET1 may play a protective role in allergic airway inflammation through activation of the AhR signaling pathways in airway epithelial cells, probably through epigenetic modifications of the AhR gene (Fig. 4b) [178].

Depending on the cell type and which TFs and histone-modifying enzymes TET proteins interact with, they can either activate or repress gene expression. For example, TET1 regulates hypoxia-induced epithelial-mesenchymal transition by acting as a co-activator [179]. In embryonic stem cells with catalytically inactive Tet1 (*Tet1*^{m/m}) and Tet1 knockout (*Tet1*^{-/-}), loss of Tet1, rather than its catalytic activity, led to abnormal upregulation of bivalent developmental genes, involving chromatin modifiers like SIN3A and reducing H3K27 trimethylation and deacetylation without changes in DNA methylation. *Tet1*^{-/-} embryos expressed elevated levels of *Gata6* and experienced developmental delays, highlighting the crucial non-catalytic roles of TET1 in regulating H3K27 modifications in gene silencing [180]. Especially relevant to wildfire smoke exposure, AhR and TET proteins have a complex regulatory relationship in hematopoietic and lymphoid cells, in particular, in modifying cell differentiation and lineage specificity during inflammation

[181]. For example, AhR directly upregulates *Tet2* expression (reviewed in [182]). TET2 is involved in promoting plasticity rather than differentiation of hematopoietic cells leading to malignancies and chronic inflammatory conditions (reviewed in [183]). This is important as during wildfire smoke exposure TET2 can enable AhR activation of genes such as *Il17a* leading to plasticity (Fig. 4c) [177,184]. The role of TET proteins and their interactions with AhR in wildfire smoke and other toxic exposures warrant further clarifications.

Conclusions

We highlighted the mechanisms of how the harmful components PM_{2.5}, VOCs, O₃, endotoxin, microbes, and toxic gases in wildfire smoke may contribute to oxidative stress and inflammation in the airways by disrupting the respiratory tract's immune and barrier functions increasing susceptibility to allergies, asthma, and COPD. Oxidative stress, both from inhaled PM and endogenously from inflammatory cells, triggers alarmin release, damages DNA, proteins, and cell membranes, and inactivates pulmonary antioxidant and immune protective molecules like SP-A and SP-D. The AhR expressed on immune and structural cells plays a central role in directly and indirectly mediating the smoke's effects, affecting cell differentiation, and IL-17A production, implicated in neutrophilic airway inflammation. For these reasons, glucocorticoids are less effective against smoke-induced exacerbations of chronic airway diseases. The underlying mechanisms strongly point to the role of epigenetic regulation, including DNA methylation and histone modifications. Understanding these mechanisms is crucial for developing effective prevention and treatment strategies during wildfire events, highlighting the significance of basic, mechanistic research in this area.

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References

1. Henry S, Ospina MB, Dennett L, Hicks A. Assessing the risk of respiratory-related healthcare visits associated with wildfire smoke exposure in children 0-18 years old: a systematic review. *Int J Environ Res Public Health*. 2021(16):8799. [PubMed: 34444546]
2. Paudel B, Chu T, Chen M, Sampath V, Prunicki M, Nadeau KC. Increased duration of pollen and mold exposure are linked to climate change. *Sci Rep*. 2021;11(1):12816. [PubMed: 34140579]
3. Beyene T, Harvey ES, Van Buskirk J, McDonald VM, Jensen ME, Horvat JC, et al. Breathing fire': impact of prolonged bushfire smoke exposure in people with severe asthma. *Int J Environ Res Public Health*. 2022;19(12):7419. [PubMed: 35742668]
4. Blando J, Allen M, Galadima H, Tolson T, Akpinar-Elci M, Szklo-Coxe M. Observations of delayed changes in respiratory function among allergy clinic patients exposed to wildfire smoke. *Int J Environ Res Public Health*. 2022;19(3):1241. [PubMed: 35162264]

5. Moore LE, Oliveira A, Zhang R, Behjat L, Hicks A. Impacts of wildfire smoke and air pollution on a pediatric population with asthma: a population-based study. *Int J Environ Res Public Health*. 2023;20(3):1937. [PubMed: 36767304]
6. Noah TL, Worden CP, Rebuli ME, Jaspers I. The effects of wildfire smoke on asthma and allergy. *Curr Allergy Asthma Rep*. 2023;23(7):375–87. [PubMed: 37171670]
7. Ghio AJ. Mechanism of asthmatic exacerbation by ambient air pollution particles. *Expert Rev Respir Med*. 2008;2(1):109–18. [PubMed: 20477227]
8. Peden DB. The “envirome” and what the practitioner needs to know about it. *Ann Allergy Asthma Immunol*. 2019;123(6):542–9. [PubMed: 31560947]
9. Reid CE, Jerrett M, Tager IB, Petersen ML, Mann JK, Balmes JR. Differential respiratory health effects from the 2008 northern California wildfires: a spatiotemporal approach. *Environ Res*. 2016;150:227–35. [PubMed: 27318255]
10. Reid CE, Maestas MM. Wildfire smoke exposure under climate change: impact on respiratory health of affected communities. *Curr Opin Pulm Med*. 2019;25(2):179–87. [PubMed: 30461534]
11. Diaz J. Smoke from Canada wildfires is returning to New York. *The New York Times*. 2023;2023.
12. Kunzli N, Avol E, Wu J, Gauderman WJ, Rappaport E, Millstein J, et al. Health effects of the 2003 Southern California wildfires on children. *Am J Respir Crit Care Med*. 2006;174(11):1221–8. [PubMed: 16946126]
13. Liu JC, Pereira G, Uhl SA, Bravo MA, Bell ML. A systematic review of the physical health impacts from non-occupational exposure to wildfire smoke. *Environ Res*. 2015;136:120–32. [PubMed: 25460628]
14. Redai I, Haczku A. Air pollution and chronic obstructive airway disease. Air pollution and health effects. In: Nadadur S, Hollingsworth J, editors. *Molecular and integrative toxicology*. Molecular and integrative toxicology. London: Springer; 2015. Vol. Air Pollution and Health Effects. p. 119–49.
15. Reid CE, Jerrett M, Petersen ML, Pfister GG, Morefield PE, Tager IB, et al. Spatiotemporal prediction of fine particulate matter during the 2008 northern California wildfires using machine learning. *Environ Sci Technol*. 2015;49(6):3887–96. [PubMed: 25648639]
16. Tse K, Chen L, Tse M, Zuraw B, Christiansen S. Effect of catastrophic wildfires on asthmatic outcomes in obese children: breathing fire. *Ann Allergy Asthma Immunol*. 2015;114(4):308–11.e4. [PubMed: 25747784]
17. Reid CE, Brauer M, Johnston FH, Jerrett M, Balmes JR, Elliott CT. Critical review of health impacts of wildfire smoke exposure. *Environ Health Perspect*. 2016;124(9):1334–43. [PubMed: 27082891]
18. Vicedo-Cabrera AM, Esplugues A, Iniguez C, Estarlich M, Ballester F. Health effects of the 2012 Valencia (Spain) wildfires on children in a cohort study. *Environ Geochem Health*. 2016;38(3):703–12. [PubMed: 26215426]
19. Morgan BW, Grigsby MR, Siddharthan T, Chowdhury M, Rubinstein A, Gutierrez L, et al. Epidemiology and risk factors of asthma-chronic obstructive pulmonary disease overlap in low- and middle-income countries. *J Allergy Clin Immunol*. 2019;143(4):1598–606. [PubMed: 30291842]
20. Reid CE, Considine EM, Watson GL, Telesca D, Pfister GG, Jerrett M. Associations between respiratory health and ozone and fine particulate matter during a wildfire event. *Environ Int*. 2019;129:291–8. [PubMed: 31146163]
21. Rossiello MR, Szema A. Health effects of climate change-induced wildfires and heatwaves. *Cureus*. 2019;11(5):e4771. [PubMed: 31363452]
22. Zeki AA, Flayer CH, Haczku A. A burning need to redefine airways disease: biomass smoke exposure identified as a unique risk factor for asthma-chronic obstructive pulmonary disease overlap in low- and middle-income countries. *J Allergy Clin Immunol*. 2019;143(4):1339–41. [PubMed: 30529450]
23. Burte E, Leynaert B, Marcon A, Bousquet J, Benmerad M, Bono R, et al. Long-term air pollution exposure is associated with increased severity of rhinitis in 2 European cohorts. *J Allergy Clin Immunol*. 2020;145(3):834–42.e6. [PubMed: 31983528]

24. Aguilera R, Corringham T, Gershunov A, Benmarhnia T. Wildfire smoke impacts respiratory health more than fine particles from other sources: observational evidence from Southern California. *Nat Commun.* 2021;12(1):1493. [PubMed: 33674571]
25. Chen H, Samet JM, Bromberg PA, Tong H. Cardiovascular health impacts of wildfire smoke exposure. *Part Fibre Toxicol.* 2021;18(1):2. [PubMed: 33413506]
26. Pacheco SE, Guidos-Fogelbach G, Annesi-Maesano I, Pawankar R, D'Amato G, Latour-Staffeld P, et al. Climate change and global issues in allergy and immunology. *J Allergy Clin Immunol.* 2021;148(6):1366–77. [PubMed: 34688774]
27. Ye T, Guo Y, Chen G, Yue X, Xu R, Coelho M, et al. Risk and burden of hospital admissions associated with wildfire-related PM_{2.5} in Brazil, 2000–15: a nationwide time-series study. *Lancet Planet Health.* 2021;5(9):e599–607. [PubMed: 34508681]
28. Akdis CA, Nadeau KC. Human and planetary health on fire. *Nat Rev Immunol.* 2022;22(11):651–2. [PubMed: 36042380]
29. Faridi S, Krzyzanowski M, Cohen AJ, Malkawi M, Moh'd Safi HA, Yousefian F, et al. Ambient air quality standards and policies in eastern mediterranean countries: a review. *Int J Public Health.* 2023;68:1605352. [PubMed: 36891223]
30. Vienneau D, Stafoggia M, Rodopoulou S, Chen J, Atkinson RW, Bauwelinck M, et al. Association between exposure to multiple air pollutants, transportation noise and cause-specific mortality in adults in Switzerland. *Environ Health.* 2023;22(1):29. [PubMed: 36967400]
31. Rice MB, Henderson SB, Lambert AA, Cromar KR, Hall JA, Cascio WE, et al. Respiratory impacts of wildland fire smoke: future challenges and policy opportunities. An official American thoracic society workshop report. *Ann Am Thorac Soc.* 2021;18(6):921–30. [PubMed: 33938390]
32. NASEM NAoS. Engineering and medicine. Wildland fires: toward improved understanding and forecasting of air quality impacts: proceedings of a workshop. Washington, DC: The National Academies Press;2022.
33. OEHHA C. OEHHA acute, 8-hour and chronic reference exposure level (REL) summary as of august, 2020. In: OEHHA, editor. OEHHA acute, 8-hour and chronic reference exposure level (REL) Summary: [CA.gov](https://oehha.ca.gov/); 2020.
34. Johnston FH, Webby RJ, Pilotto LS, Bailie RS, Parry DL, Halpin SJ. Vegetation fires, particulate air pollution and asthma: a panel study in the Australian monsoon tropics. *Int J Environ Health Res.* 2006;16(6):391–404. [PubMed: 17164166]
35. Wegesser TC, Franzi LM, Mitloehner FM, Eiguren-Fernandez A, Last JA. Lung antioxidant and cytokine responses to coarse and fine particulate matter from the great California wildfires of 2008. *Inhal Toxicol.* 2010;22(7):561–70. [PubMed: 20388000]
36. Williams KM, Franzi LM, Last JA. Cell-specific oxidative stress and cytotoxicity after wildfire coarse particulate matter instillation into mouse lung. *Toxicol Appl Pharmacol.* 2013;266(1):48–55. [PubMed: 23142465]
37. Nakayama Wong LS, Aung HH, Lame MW, Wegesser TC, Wilson DW. Fine particulate matter from urban ambient and wildfire sources from California's San Joaquin Valley initiate differential inflammatory, oxidative stress, and xenobiotic responses in human bronchial epithelial cells. *Toxicol Vitro.* 2011;25(8):1895–905.
38. Franzi LM, Bratt JM, Williams KM, Last JA. Why is particulate matter produced by wildfires toxic to lung macrophages? *Toxicol Appl Pharmacol.* 2011;257(2):182–8. [PubMed: 21945489]
39. Kim YH, Tong H, Daniels M, Boykin E, Krantz QT, McGee J, et al. Cardiopulmonary toxicity of peat wildfire particulate matter and the predictive utility of precision cut lung slices. *Part Fibre Toxicol.* 2014;11:29. [PubMed: 24934158]
40. Dunn JLM, Kartchner LB, Stepp WH, Glenn LI, Malfitano MM, Jones SW, et al. Blocking CXCL1-dependent neutrophil recruitment prevents immune damage and reduces pulmonary bacterial infection after inhalation injury. *Am J Physiol Lung Cel Mol Physiol.* 2018;314(5):L822–34.
41. Woolery S, Willner J, Prahlow JA, Douglas E. Death after poison ivy smoke inhalation. *Am J Forensic Med Pathol.* 2022;43(4):359–62. [PubMed: 35703240]

42. Kobziar LN, Pingree MRA, Larson H, Dreaden TJ, Green S, Smith JA. Pyroaerobiology: the aerosolization and transport of viable microbial life by wildland fire. *Ecosphere*. 2018;9(11):e02507.
43. Kobziar LN, Vuono D, Moore R, Christner BC, Dean T, Betancourt D, et al. Wildland fire smoke alters the composition, diversity, and potential atmospheric function of microbial life in the aerobiome. *ISME Commun*. 2022;2(1):8. [PubMed: 37938277]
44. Kunovac A, Hathaway QA, Pinti MV, Taylor AD, Hollander JM. Cardiovascular adaptations to particle inhalation exposure: molecular mechanisms of the toxicology. *Am J Physiol Heart Circ Physiol*. 2020;319(2):H282–305. [PubMed: 32559138]
45. Hill W, Lim EL, Weeden CE, Lee C, Augustine M, Chen K, et al. Lung adenocarcinoma promotion by air pollutants. *Nature*. 2023;616(7955):159–67. [PubMed: 37020004]
46. Vogel CFA, Van Winkle LS, Esser C, Haarmann-Stemmann T. The aryl hydrocarbon receptor as a target of environmental stressors - implications for pollution mediated stress and inflammatory responses. *Redox Biol*. 2020;34:101530. [PubMed: 32354640]
47. Wegesser TC, Pinkerton KE, Last JA. California wildfires of 2008: coarse and fine particulate matter toxicity. *Environ Health Perspect*. 2009;117(6):893–7. [PubMed: 19590679]
48. Urbanski SP, Hao WM, Baker S. Chemical composition of wildland fire emissions. In: Bytnerowicz A, Arbaugh M, Andersen C, Riebau A, editors. *Wildland fires and air pollution developments in environmental science 8*. Amsterdam, The Netherlands: Elsevier; 2009. p. 79–108.
49. Rager JE, Clark J, Eaves LA, Avula V, Niehoff NM, Kim YH, et al. Mixtures modeling identifies chemical inducers versus repressors of toxicity associated with wildfire smoke. *Sci Total Environ*. 2021;775:145759. [PubMed: 33611182]
50. Tiotiu AI, Novakova P, Nedeva D, Chong-Neto HJ, Novakova S, Steiropoulos P, et al. Impact of air pollution on asthma outcomes. *Int J Environ Res Public Health*. 2020;17(17):6212. [PubMed: 32867076]
51. Reinhardt TE, Ottmar RD. Baseline measurements of smoke exposure among wildland firefighters. *J Occup Environ Hyg*. 2004;1(9):593–606. [PubMed: 15559331]
52. Kim KH, Jahan SA, Kabir E. A review on human health perspective of air pollution with respect to allergies and asthma. *Environ Int*. 2013;59:41–52. [PubMed: 23770580]
53. Liu N, Bu Z, Liu W, Kan H, Zhao Z, Deng F, et al. Health effects of exposure to indoor volatile organic compounds from 1980 to 2017: a systematic review and meta-analysis. *Indoor Air*. 2022;32(5):e13038. [PubMed: 35622720]
54. Rufo JC, Annesi-Maesano I, Carreiro-Martins P, Moreira A, Sousa AC, Pastorinho MR, et al. Issue 2 - “Update on adverse respiratory effects of indoor air pollution” Part 1): indoor air pollution and respiratory diseases: a general update and a Portuguese perspective. *Pulmonology*. 2023.
55. Lu C, Wang F, Liu Q, Deng M, Yang X, Ma P. Effect of NO(2) exposure on airway inflammation and oxidative stress in asthmatic mice. *J Hazard Mater*. 2023;457:131787. [PubMed: 37295329]
56. de Souza Fernandes Duarte E, Salgueiro V, Costa MJ, Lucio PS, Potes M, Bortoli D, et al. Fire-pollutant-atmosphere components and its impact on mortality in Portugal during wildfire seasons. *Geohealth*. 2023;7(10):e2023GH000802.
57. Song J, Qiu W, Huang X, Guo Y, Chen W, Wang D, et al. Association of ambient carbon monoxide exposure with hospitalization risk for respiratory diseases: a time series study in Ganzhou, China. *Front Public Health*. 2023;11:1106336. [PubMed: 36866098]
58. Poole JA, Barnes CS, Demain JG, Bernstein JA, Padukudru MA, Sheehan WJ, et al. Impact of weather and climate change with indoor and outdoor air quality in asthma: a work group report of the aaaa environmental exposure and respiratory health committee. *J Allergy Clin Immunol*. 2019;143(5):1702–10. [PubMed: 30826366]
59. Jerrett M, Burnett RT, Pope CA 3rd, Ito K, Thurston G, Krewski D, et al. Long-term ozone exposure and mortality. *N Engl J Med*. 2009;360(11):1085–95. [PubMed: 19279340]
60. Rosser F, Balmes J. Ozone and childhood respiratory health: a primer for US pediatric providers and a call for a more protective standard. *Pediatr Pulmonol*. 2023;58(5):1355–66. [PubMed: 36815617]

61. Enweasor C, Flayer CH, Haczku A. Ozone-induced oxidative stress, neutrophilic airway inflammation, and glucocorticoid resistance in asthma. *Front Immunol.* 2021;12:631092. [PubMed: 33717165]
62. Michalec L, Choudhury BK, Postlethwait E, Wild JS, Alam R, Lett-Brown M, et al. CCL7 and CXCL10 orchestrate oxidative stress-induced neutrophilic lung inflammation. *J Immunol.* 2002;168(2):846–52. [PubMed: 11777981]
63. Johnston RA, Schwartzman IN, Flynt L, Shore SA. Role of interleukin-6 in murine airway responses to ozone. *Am J Physiol Lung Cel Mol Physiol.* 2005;288(2):L390–7.
64. Kierstein S, Poulain FR, Cao Y, Grous M, Mathias R, Kierstein G, et al. Susceptibility to ozone-induced airway inflammation is associated with decreased levels of surfactant protein D. *Respir Res.* 2006;7(1):85. [PubMed: 16740162]
65. Cho HY, Kleeberger SR. Genetic mechanisms of susceptibility to oxidative lung injury in mice. *Free Radic Biol Med.* 2007;42(4):433–45. [PubMed: 17275675]
66. Wiegman CH, Li F, Clarke CJ, Jazrawi E, Kirkham P, Barnes PJ, et al. A comprehensive analysis of oxidative stress in the ozone-induced lung inflammation mouse model. *Clin Sci.* 2014;126(6):425–40.
67. Ge MQ, Kokalari B, Flayer CH, Killingbeck SS, Redai IG, MacFarlane AW, et al. Cutting edge: role of NK cells and surfactant protein D in dendritic cell lymph node homing: effects of ozone exposure. *J Immunol.* 2016;196(2):553–7. [PubMed: 26673133]
68. Yang Q, Ge MQ, Kokalari B, Redai IG, Wang X, Kemeny DM, et al. Group 2 innate lymphoid cells mediate ozone-induced airway inflammation and hyperresponsiveness in mice. *J Allergy Clin Immunol.* 2016;137(2):571–8. [PubMed: 26282284]
69. Bao A, Yang H, Ji J, Chen Y, Bao W, Li F, et al. Involvements of p38 MAPK and oxidative stress in the ozone-induced enhancement of AHR and pulmonary inflammation in an allergic asthma model. *Respir Res.* 2017;18(1):216. [PubMed: 29284473]
70. Yousefi S, Sharma SK, Stojkov D, Germic N, Aeschlimann S, Ge MQ, et al. Oxidative damage of SP-D abolishes control of eosinophil extracellular DNA trap formation. *J Leukoc Biol.* 2018;104(1):205–14. [PubMed: 29733456]
71. Flayer CH, Ge MQ, Hwang JW, Kokalari B, Redai IG, Jiang Z, et al. Ozone inhalation attenuated the effects of budesonide on *Aspergillus fumigatus*-induced airway inflammation and hyperreactivity in mice. *Front Immunol.* 2019;10:2173. [PubMed: 31572383]
72. Michaudel C, Bataille F, Maillet I, Fauconnier L, Colas C, Sokol H, et al. Ozone-induced aryl hydrocarbon receptor activation controls lung inflammation via interleukin-22 modulation. *Front Immunol.* 2020;11:144. [PubMed: 32161582]
73. Xue Y, Bao W, Zhou Y, Fu Q, Hao H, Han L, et al. Small-airway dysfunction is involved in the pathogenesis of asthma: evidence from two mouse models. *J Asthma Allergy.* 2021;14:883–96. [PubMed: 34285515]
74. Johnston RA, Atkins CL, Siddiqui SR, Jackson WT, Mitchell NC, Spencer CY, et al. Interleukin-11 receptor subunit α -1 is required for maximal airway responsiveness to methacholine after acute exposure to ozone. *Am J Physiol Regul Integr Comp Physiol.* 2022;323(6):R921–34. [PubMed: 36283092]
75. Ho K, Weimar D, Torres-Matias G, Lee H, Shamsi S, Shalosky E, et al. Ozone impairs endogenous compensatory responses in allergic asthma. *Toxicol Appl Pharmacol.* 2023;459:116341. [PubMed: 36502870]
76. Hyde DM, Hubbard WC, Wong V, Wu R, Pinkerton K, Plopper CG. Ozone-induced acute tracheobronchial epithelial injury: relationship to granulocyte emigration in the lung. *Am J Respir Cel Mol Biol.* 1992;6(5):481–97.
77. Murphy SR, Schelegle ES, Edwards PC, Miller LA, Hyde DM, Van Winkle LS. Postnatal exposure history and airways: oxidant stress responses in airway explants. *Am J Respir Cel Mol Biol.* 2012;47(6):815–23.
78. Flayer CH, Larson ED, Joseph A, Kao S, Qu W, Van Haren A, et al. Ozone-induced enhancement of airway hyperreactivity in rhesus macaques: effects of antioxidant treatment. *J Allergy Clin Immunol.* 2020;145(1):312–23. [PubMed: 31627909]

79. Royer C, Miller LA, Haczku A. A novel nonhuman primate model of nonatopic asthma. *Methods Mol Biol.* 2022;2506:83–94. [PubMed: 35771465]
80. Horvath SM, Gliner JA, Folinsbee LJ. Adaptation to ozone: duration of effect. *Am Rev Respir Dis.* 1981;123(5):496–9. [PubMed: 7235372]
81. Alexis NE, Lay JC, Haczku A, Gong H, Linn W, Hazucha MJ, et al. Fluticasone propionate protects against ozone-induced airway inflammation and modified immune cell activation markers in healthy volunteers. *Environ Health Perspect.* 2008;116(6):799–805. [PubMed: 18560537]
82. Alexis NE, Lay JC, Hazucha M, Harris B, Hernandez ML, Bromberg PA, et al. Low-level ozone exposure induces airways inflammation and modifies cell surface phenotypes in healthy humans. *Inhal Toxicol.* 2010;22(7):593–600. [PubMed: 20384440]
83. Kim CS, Alexis NE, Rappold AG, Kehrl H, Hazucha MJ, Lay JC, et al. Lung function and inflammatory responses in healthy young adults exposed to 0.06 ppm ozone for 6.6 hours. *Am J Respir Crit Care Med.* 2011;183(9):1215–21. [PubMed: 21216881]
84. Vagaggini B, Taccola M, Cianchetti S, Carnevali S, Bartoli ML, Bacci E, et al. Ozone exposure increases eosinophilic airway response induced by previous allergen challenge. *Am J Respir Crit Care Med.* 2002;166(8):1073–7. [PubMed: 12379550]
85. Hernandez ML, Lay JC, Harris B, Esther CR Jr, Brickey WJ, Bromberg PA, et al. Atopic asthmatic subjects but not atopic subjects without asthma have enhanced inflammatory response to ozone. *J Allergy Clin Immunol.* 2010;126(3):537–44 e1. [PubMed: 20816188]
86. Kirkham PA, Caramori G, Casolari P, Papi AA, Edwards M, Shamji B, et al. Oxidative stress-induced antibodies to carbonyl-modified protein correlate with severity of chronic obstructive pulmonary disease. *Am J Respir Crit Care Med.* 2011;184(7):796–802. [PubMed: 21965015]
87. Peden DB. The role of oxidative stress and innate immunity in O₃ and endotoxin-induced human allergic airway disease. *Immunol Rev.* 2011;242(1):91–105. [PubMed: 21682740]
88. Moore BD, Hyde D, Miller L, Wong E, Frelinger J, Schelegle ES. Allergen and ozone exacerbate serotonin-induced increases in airway smooth muscle contraction in a model of childhood asthma. *Respiration.* 2012;83(6):529–42. [PubMed: 22507883]
89. Alexis NE, Carlsten C. Interplay of air pollution and asthma immunopathogenesis: a focused review of diesel exhaust and ozone. *Int Immunopharmacol.* 2014;23(1):347–55. [PubMed: 25194677]
90. Moore BD, Hyde DM, Miller LA, Wong EM, Schelegle ES. Persistence of serotonergic enhancement of airway response in a model of childhood asthma. *Am J Respir Cell Mol Biol.* 2014;51(1):77–85.
91. EPA USEPA. Trends in ozone adjusted for weather conditions. *Air Trends.* 2022.
92. Malashock DA, Delang MN, Becker JS, Serre ML, West JJ, Chang KL, et al. Global trends in ozone concentration and attributable mortality for urban, peri-urban, and rural areas between 2000 and 2019: a modelling study. *Lancet Planet Health.* 2022;6(12):e958–67. [PubMed: 36495890]
93. Aguilera J, Kaushik A, Cauwenberghs N, Heider A, Ogulur I, Yazici D, et al. Granzymes, IL-16, and poly(ADP-ribose) polymerase 1 increase during wildfire smoke exposure. *J Allergy Clin Immunol Glob.* 2023;2(2):100093. [PubMed: 37539354]
94. Parron-Ballesteros J, Gordo RG, Lopez-Rodriguez JC, Olmo N, Villalba M, Batanero E, et al. Beyond allergic progression: from molecules to microbes as barrier modulators in the gut-lung axis functionality. *Front Allergy.* 2023;4:1093800. [PubMed: 36793545]
95. Roscioli E, Hamon R, Lester SE, Jersmann HPA, Reynolds PN, Hodge S. Airway epithelial cells exposed to wildfire smoke extract exhibit dysregulated autophagy and barrier dysfunction consistent with COPD. *Respir Res.* 2018;19(1):234. [PubMed: 30486816]
96. Bekki K, Ito T, Yoshida Y, He C, Arashidani K, He M, et al. PM_{2.5} collected in China causes inflammatory and oxidative stress responses in macrophages through the multiple pathways. *Environ Toxicol Pharmacol.* 2016;45:362–9. [PubMed: 27393915]
97. da Costa Loureiro L, da Costa Loureiro L, Gabriel-Junior EA, Zambuzi FA, Fontanari C, Sales-Campos H, et al. Pulmonary surfactant phosphatidylcholines induce immunological adaptation of alveolar macrophages. *Mol Immunol.* 2020;122:163–72. [PubMed: 32361419]
98. Kelly FJ, Mudway I, Krishna MT, Holgate ST. The free radical basis of air pollution: focus on ozone. *Respir Med.* 1995;89(10):647–56. [PubMed: 8570878]

99. Bromberg PA. Mechanisms of the acute effects of inhaled ozone in humans. *Biochim Biophys Acta*. 2016;1860(12):2771–81. [PubMed: 27451958]
100. Girotti AW. Lipid hydroperoxide generation, turnover, and effector action in biological systems. *J Lipid Res*. 1998;39(8):1529–42. [PubMed: 9717713]
101. Szabo C, Ischiropoulos H, Radi R. Peroxynitrite: biochemistry, pathophysiology and development of therapeutics. *Nat Rev Drug Discov*. 2007;6(8):662–80. [PubMed: 17667957]
102. Sahiner UM, Birben E, Erzurum S, Sackesen C, Kalayci O. Oxidative stress in asthma: Part of the puzzle. *Pediatr Allergy Immunol*. 2018;29(8):789–800. [PubMed: 30069955]
103. Barnes PJ. Reactive oxygen species and airway inflammation. *Free Radic Biol Med*. 1990;9(3):235–43. [PubMed: 2272532]
104. Barnes PJ. Oxidative stress in chronic obstructive pulmonary disease. *Antioxidants*. 2022;11(5):965. [PubMed: 35624831]
105. Erzurum SC. New insights in oxidant biology in asthma. *Ann Am Thorac Soc*. 2016;13(Suppl 1):S35–9. [PubMed: 27027950]
106. Dweik RA, Boggs PB, Erzurum SC, Irvin CG, Leigh MW, Lundberg JO, et al. An official ATS clinical practice guideline: interpretation of exhaled nitric oxide levels (FENO) for clinical applications. *Am J Respir Crit Care Med*. 2011;184(5):602–15. [PubMed: 21885636]
107. Brand JD, Mathews JA, Kasahara DI, Wurmbrand AP, Shore SA. Regulation of IL-17A expression in mice following sub-acute ozone exposure. *J Immunotoxicol*. 2016;13(3):428–38. [PubMed: 27043160]
108. Mathews JA, Krishnamoorthy N, Kasahara DI, Cho Y, Wurmbrand AP, Ribeiro L, et al. IL-33 drives augmented responses to ozone in obese mice. *Environ Health Perspect*. 2017;125(2):246–53. [PubMed: 27472835]
109. Shore SA. Mechanistic basis for obesity-related increases in ozone-induced airway hyperresponsiveness in mice. *Ann Am Thorac Soc*. 2017;14(Supplement_5):S357–62. [PubMed: 29161088]
110. Michaudel C, Mackowiak C, Maillet I, Fauconnier L, Akdis CA, Sokolowska M, et al. Ozone exposure induces respiratory barrier biphasic injury and inflammation controlled by IL-33. *J Allergy Clin Immunol*. 2018;142(3):942–58. [PubMed: 29331644]
111. Kasahara DI, Wilkinson JE, Cho Y, Cardoso AP, Huttenhower C, Shore SA. The interleukin-33 receptor contributes to pulmonary responses to ozone in male mice: role of the microbiome. *Respir Res*. 2019;20(1):197. [PubMed: 31455422]
112. Mumby S, Chung KF, Adcock IM. Transcriptional effects of ozone and impact on airway inflammation. *Front Immunol*. 2019;10:1610. [PubMed: 31354743]
113. Kasahara DI, Shore SA. IL-33, diet-induced obesity, and pulmonary responses to ozone. *Respir Res*. 2020;21(1):98. [PubMed: 32326950]
114. Wiegman CH, Li F, Ryffel B, Togbe D, Chung KF. Oxidative stress in ozone-induced chronic lung inflammation and emphysema: a facet of chronic obstructive pulmonary disease. *Front Immunol*. 2020;11:1957. [PubMed: 32983127]
115. Black C, Gerriets JE, Fontaine JH, Harper RW, Kenyon NJ, Tablin F, et al. Early life wildfire smoke exposure is associated with immune dysregulation and lung function decrements in adolescence. *Am J Respir Cell Mol Biol*. 2017;56(5):657–66.
116. Kierstein S, Krytska K, Sharma S, Amrani Y, Salmon M, Panettieri RA Jr, et al. Ozone inhalation induces exacerbation of eosinophilic airway inflammation and hyperresponsiveness in allergen-sensitized mice. *Allergy*. 2008;63(4):438–46. [PubMed: 18315731]
117. Bontinck A, Maes T, Joos G. Asthma and air pollution: recent insights in pathogenesis and clinical implications. *Curr Opin Pulm Med*. 2020;26(1):10–9. [PubMed: 31724961]
118. Deterding RR, Shimizu H, Fisher JH, Shannon JM. Regulation of surfactant protein D expression by glucocorticoids in vitro and in vivo. *Am J Respir Cell Mol Biol*. 1994;10(1):30–7.
119. Cao Y, Tao JQ, Bates SR, Beers MF, Haczk A. IL-4 induces production of the lung collectin surfactant protein-D. *J Allergy Clin Immunol*. 2004;113(3):439–44. [PubMed: 15007344]
120. Yousefi S, Sharma SK, Flayer CH, Stojkov D, Ge MQ, Germic N, et al. Oxidative damage abolishes the ability of SP-D to block eosinophil extracellular DNA trap release during exacerbation of allergic airway inflammation. *Am J Resp Crit Care*. 2018;197.

121. Yousefi S, Qu W, Flayer CH, Stojkov D, Germic N, Aeschlimann S, et al. Surfactant protein D (SP-D) inhibits neutrophil extracellular DNA trap formation: effects of S-nitrosylation. *J Allergy Clin Immunol.* 2019;143(2):Ab192–Ab92.
122. Stevens WH, Adelroth E, Wattie J, Woolley MJ, Ellis R, Dahlback M, et al. Effect of inhaled budesonide on ozone-induced airway hyperresponsiveness and bronchoalveolar lavage cells in dogs. *J Appl Physiol.* 1994;77(6):2578–83. [PubMed: 7896594]
123. Joad JP, Kott KS, Bric JM, Schelegle ES, Gershwin LJ, Plopper CG, et al. The effects of inhaled corticosteroids on intrinsic responsiveness and histology of airways from infant monkeys exposed to house dust mite allergen and ozone. *Toxicol Appl Pharmacol.* 2008;226(2):153–60. [PubMed: 17991502]
124. Vagaggini B, Cianchetti S, Bartoli M, Ricci M, Bacci E, Dente FL, et al. Prednisone blunts airway neutrophilic inflammatory response due to ozone exposure in asthmatic subjects. *Respiration.* 2007;74(1):61–8. [PubMed: 17028419]
125. Bruijnzeel PL, Uddin M, Koenderman L. Targeting neutrophilic inflammation in severe neutrophilic asthma: can we target the disease-relevant neutrophil phenotype? *J Leukoc Biol.* 2015;98(4):549–56. [PubMed: 25977288]
126. Poulain-Godefroy O, Boute M, Carrard J, Alvarez-Simon D, Tsicopoulos A, de Nadai P. The aryl hydrocarbon receptor in asthma: friend or foe? *Int J Mol Sci.* 2020;21(22):8797. [PubMed: 33233810]
127. Beamer CA, Shepherd DM. Role of the aryl hydrocarbon receptor (AhR) in lung inflammation. *Semin Immunopathol.* 2013;35(6):693–704. [PubMed: 23963493]
128. Gutierrez-Vazquez C, Quintana FJ. Regulation of the immune response by the aryl hydrocarbon receptor. *Immunity.* 2018;48(1):19–33. [PubMed: 29343438]
129. Jaligama S, Patel VS, Wang P, Sallam A, Harding J, Kelley M, et al. Radical containing combustion derived particulate matter enhance pulmonary Th17 inflammation via the aryl hydrocarbon receptor. *Part Fibre Toxicol.* 2018;15(1):20. [PubMed: 29724254]
130. Harding JN, Gross M, Patel V, Potter S, Cormier SA. Association between particulate matter containing EPFRs and neutrophilic asthma through AhR and Th17. *Respir Res.* 2021;22(1):275. [PubMed: 34702270]
131. Nurieva R, Yang XO, Martinez G, Zhang Y, Panopoulos AD, Ma L, et al. Essential autocrine regulation by IL-21 in the generation of inflammatory T cells. *Nature.* 2007;448(7152):480–3. [PubMed: 17581589]
132. Mathews JA, Williams AS, Brand JD, Wurmbrand AP, Chen L, Ninin FM, et al. $\gamma\delta$ T cells are required for pulmonary IL-17A expression after ozone exposure in mice: role of TNF α . *PLoS One.* 2014;9(5):e97707. [PubMed: 24823369]
133. Chesne J, Braza F, Mahay G, Brouard S, Aronica M, Magnan A. IL-17 in severe asthma. Where do we stand? *Am J Respir Crit Care Med.* 2014;190(10):1094–101. [PubMed: 25162311]
134. Haczku A, Panettieri RA Jr. Social stress and asthma: the role of corticosteroid insensitivity. *J Allergy Clin Immunol.* 2010;125(3):550–8. [PubMed: 20153032]
135. Barnes PJ. Glucocorticosteroids. *Handb Exp Pharmacol.* 2017;237:93–115. [PubMed: 27796513]
136. Khantakova JN, Mutovina A, Ayriyants KA, Bondar NP. Th17 cells, glucocorticoid resistance, and depression. *Cells.* 2023;12(23):2749. [PubMed: 38067176]
137. Gaffen S. IL-17 receptor composition. *Nat Rev Immunol.* 2016;16(1):4.
138. Amatya N, Garg AV, Gaffen SL. IL-17 signaling: the yin and the yang. *Trends Immunol.* 2017;38(5):310–22. [PubMed: 28254169]
139. Amatya N, Childs EE, Cruz JA, Aggor FEY, Garg AV, Berman AJ, et al. IL-17 integrates multiple self-reinforcing, feed-forward mechanisms through the RNA binding protein Arid5a. *Sci Signal.* 2018;11(551):11.
140. Li X, Bechara R, Zhao J, McGeachy MJ, Gaffen SL. IL-17 receptor-based signaling and implications for disease. *Nat Immunol.* 2019;20(12):1594–602. [PubMed: 31745337]
141. Barnes PJ. Corticosteroid resistance in patients with asthma and chronic obstructive pulmonary disease. *J Allergy Clin Immunol.* 2013;131(3):636–45. [PubMed: 23360759]

142. McKinley L, Alcorn JF, Peterson A, Dupont RB, Kapadia S, Logar A, et al. TH17 cells mediate steroid-resistant airway inflammation and airway hyperresponsiveness in mice. *J Immunol.* 2008;181(6):4089–97. [PubMed: 18768865]
143. Bulygin AS, Khantakova JN, Shkaruba NS, Shiku H, Sennikov SS. The role of metabolism on regulatory T cell development and its impact in tumor and transplantation immunity. *Front Immunol.* 2022;13:1016670. [PubMed: 36569866]
144. Bollati V, Baccarelli A, Hou L, Bonzini M, Fustinoni S, Cavallo D, et al. Changes in DNA methylation patterns in subjects exposed to low-dose benzene. *Cancer Res.* 2007;67(3):876–80. [PubMed: 17283117]
145. Bellavia A, Urch B, Speck M, Brook RD, Scott JA, Albetti B, et al. DNA hypo-methylation, ambient particulate matter, and increased blood pressure: findings from controlled human exposure experiments. *J Am Heart Assoc.* 2013;2(3):e000212. [PubMed: 23782920]
146. Rossnerova A, Tulupova E, Tabashidze N, Schmuczerova J, Dostal M, Rossner P Jr, et al. Factors affecting the 27K DNA methylation pattern in asthmatic and healthy children from locations with various environments. *Mutat Res.* 2013;741-742:18–26. [PubMed: 23458556]
147. Basilio E, Chen R, Fernandez AC, Padula AM, Robinson JF, Gaw SL. Wildfire smoke exposure during pregnancy: a review of potential mechanisms of placental toxicity, impact on obstetric outcomes, and strategies to reduce exposure. *Int J Environ Res Public Health.* 2022;19(21):13727. [PubMed: 36360613]
148. Brown AP, Cai L, Laufer BI, Miller LA, LaSalle JM, Ji H. Long-term effects of wildfire smoke exposure during early life on the nasal epigenome in rhesus macaques. *Environ Int.* 2022;158:106993. [PubMed: 34991254]
149. Carberry CK, Koval LE, Payton A, Hartwell H, Ho Kim Y, Smith GJ, et al. Wildfires and extracellular vesicles: exosomal MicroRNAs as mediators of cross-tissue cardiopulmonary responses to biomass smoke. *Environ Int.* 2022;167:107419. [PubMed: 35863239]
150. Shukla A, Bunkar N, Kumar R, Bhargava A, Tiwari R, Chaudhury K, et al. Air pollution associated epigenetic modifications: transgenerational inheritance and underlying molecular mechanisms. *Sci Total Environ.* 2019;656:760–77. [PubMed: 30530146]
151. Wang C, Wang D, Zhao H, Wang J, Liu N, Shi H, et al. Traffic-related PM_{2.5} and diverse constituents disturb the balance of Th17/Treg cells by STAT3/ROR γ t-STAT5/Foxp3 signaling pathway in a rat model of asthma. *Int Immunopharmacol.* 2021;96:107788. [PubMed: 34162152]
152. Zhou Q, Tian Y, Xu C, Wang J, Jin Y. Prenatal and postnatal traffic pollution exposure, DNA methylation in Shank3 and MeCP2 promoter regions, H3K4me3 and H3K27me3 and sociability in rats' offspring. *Clin Epigenetics.* 2021;13(1):180. [PubMed: 34565458]
153. Wang C, Wang J, Zheng X, Zhang J, Zhang J, Qiao G, et al. Epigenetic regulation is involved in traffic-related PM(2.5) aggravating allergic airway inflammation in rats. *Clin Immunol.* 2022;234:108914. [PubMed: 34954131]
154. Baccarelli A, Wright RO, Bollati V, Tarantini L, Litonjua AA, Suh HH, et al. Rapid DNA methylation changes after exposure to traffic particles. *Am J Respir Crit Care Med.* 2009;179(7):572–8. [PubMed: 19136372]
155. Panni T, Mehta AJ, Schwartz JD, Baccarelli AA, Just AC, Wolf K, et al. Genome-wide analysis of DNA methylation and fine particulate matter air pollution in three study populations: KORA F3, KORA F4, and the normative aging study. *Environ Health Perspect.* 2016;124(7):983–90. [PubMed: 26731791]
156. Jiang R, Jones MJ, Sava F, Kobor MS, Carlsten C. Short-term diesel exhaust inhalation in a controlled human crossover study is associated with changes in DNA methylation of circulating mononuclear cells in asthmatics. *Part Fibre Toxicol.* 2014;11:71. [PubMed: 25487561]
157. Ding R, Jin Y, Liu X, Ye H, Zhu Z, Zhang Y, et al. Dose- and time- effect responses of DNA methylation and histone H3K9 acetylation changes induced by traffic-related air pollution. *Sci Rep.* 2017;7:43737. [PubMed: 28256616]
158. Rakyan VK, Down TA, Balding DJ, Beck S. Epigenome-wide association studies for common human diseases. *Nat Rev Genet.* 2011;12(8):529–41. [PubMed: 21747404]

159. Houseman EA, Accomando WP, Koestler DC, Christensen BC, Marsit CJ, Nelson HH, et al. DNA methylation arrays as surrogate measures of cell mixture distribution. *BMC Bioinformatics*. 2012;13:86. [PubMed: 22568884]
160. Bruniquel D, Schwartz RH. Selective, stable demethylation of the interleukin-2 gene enhances transcription by an active process. *Nat Immunol*. 2003;4(3):235–40. [PubMed: 12548284]
161. Liu J, Ballaney M, Al-alem U, Quan C, Jin X, Perera F, et al. Combined inhaled diesel exhaust particles and allergen exposure alter methylation of T helper genes and IgE production in vivo. *Toxicol Sci*. 2008;102(1):76–81. [PubMed: 18042818]
162. Zhong J, Karlsson O, Wang G, Li J, Guo Y, Lin X, et al. B vitamins attenuate the epigenetic effects of ambient fine particles in a pilot human intervention trial. *Proc Natl Acad Sci USA*. 2017;114:3503–8. [PubMed: 28289216]
163. Lin VW, Baccarelli AA, Burris HH. Epigenetics-a potential mediator between air pollution and preterm birth. *Environ Epigenet*. 2016;2(1):dvv008. [PubMed: 26900485]
164. Zhang X, Chen X, Weirauch MT, Zhang X, Burleson JD, Brandt EB, et al. Diesel exhaust and house dust mite allergen lead to common changes in the airway methylome and hydroxymethylome. *Environ Epigenet*. 2018;4(3):dvy020. [PubMed: 30090644]
165. Huang SK, Tripathi P, Koneva LA, Cavalcante RG, Craig N, Scruggs AM, et al. Effect of concentration and duration of particulate matter exposure on the transcriptome and DNA methylome of bronchial epithelial cells. *Environ Epigenet*. 2021;7(1):dvaa022. [PubMed: 33692908]
166. Sominen HK, Zhang X, Biagini Myers JM, Kovacic MB, Ulm A, Jurcak N, et al. Ten-eleven translocation 1 (TET1) methylation is associated with childhood asthma and traffic-related air pollution. *J Allergy Clin Immunol*. 2016;137(3):797–805 e5. [PubMed: 26684294]
167. Sordillo JE, Cardenas A, Qi C, Rifas-Shiman SL, Coull B, Luttmann-Gibson H, et al. Residential PM(2.5) exposure and the nasal methylome in children. *Environ Int*. 2021;153:106505. [PubMed: 33872926]
168. Gondalia R, Baldassari A, Holliday KM, Justice AE, Mendez-Giraldez R, Stewart JD, et al. Methylome-wide association study provides evidence of particulate matter air pollution-associated DNA methylation. *Environ Int*. 2019;132:104723. [PubMed: 31208937]
169. Lee MK, Xu CJ, Carnes MU, Nichols CE, Ward JM; BIOS consortium, et al. Genome-wide DNA methylation and long-term ambient air pollution exposure in Korean adults. *Clin Epigenetics*. 2019;11(1):37. [PubMed: 30819252]
170. Murphy TM, Wong CC, Arseneault L, Burrage J, Macdonald R, Hannon E, et al. Methylomic markers of persistent childhood asthma: a longitudinal study of asthma-discordant monozygotic twins. *Clin Epigenetics*. 2015;7:130. [PubMed: 26691723]
171. Vercelli D. Does epigenetics play a role in human asthma? *Allergol Int*. 2016;65(2):123–6. [PubMed: 26778244]
172. Reese SE, Xu CJ, den Dekker HT, Lee MK, Sikdar S, Ruiz-Arenas C, et al. Epigenome-wide meta-analysis of DNA methylation and childhood asthma. *J Allergy Clin Immunol*. 2019;143(6):2062–74. [PubMed: 30579849]
173. Xu CJ, Gruzdeva O, Qi C, Esplugues A, Gehring U, Bergstrom A, et al. Shared DNA methylation signatures in childhood allergy: the MeDALL study. *J Allergy Clin Immunol*. 2021;147(3):1031–40. [PubMed: 33338541]
174. Krauss-Etschmann S, Meyer KF, Dehmel S, Hylkema MN. Inter- and transgenerational epigenetic inheritance: evidence in asthma and COPD? *Clin Epigenetics*. 2015;7(1):53. [PubMed: 26052354]
175. Zong DD, Ouyang RY, Chen P. Epigenetic mechanisms in chronic obstructive pulmonary disease. *Eur Rev Med Pharmacol Sci*. 2015;19(5):844–56. [PubMed: 25807439]
176. Li D, Chen J, Pei D. The battle between TET proteins and DNA methylation for the right cell. *Trends Cel Biol*. 2018;28(12):973–5.
177. Ji H, Brown AP, Henson SN, Haczku A. The “epiTet” of air pollution: epigenetic regulation of airway inflammation by Tet1. *Int Arch Allergy Immunol*. 2023;184(9):949–52. [PubMed: 37276849]

178. Zhong J, Li X, Cai W, Wang Y, Dong S, Yang J, et al. TET1 modulates H4K16 acetylation by controlling auto-acetylation of hMOF to affect gene regulation and DNA repair function. *Nucleic Acids Res.* 2017;45(2):672–84. [PubMed: 27733505]
179. Tsai YP, Chen HF, Chen SY, Cheng WC, Wang HW, Shen ZJ, et al. TET1 regulates hypoxia-induced epithelial-mesenchymal transition by acting as a co-activator. *Genome Biol.* 2014;15(12):513. [PubMed: 25517638]
180. Chrysanthou S, Tang Q, Lee J, Taylor SJ, Zhao Y, Steidl U, et al. The DNA dioxygenase Tet1 regulates H3K27 modification and embryonic stem cell biology independent of its catalytic activity. *Nucleic Acids Res.* 2022;50(6):3169–89. [PubMed: 35150568]
181. Li H, Ryu MH, Orach J, Yuen A, Lau KSK, Yeung C, et al. Acute air pollution exposure increases TET enzymes in human PBMCs. *J Allergy Clin Immunol.* 2022;150(2):477–88.e9. [PubMed: 35367469]
182. Opitz CA, Holfelder P, Prentzell MT, Trump S. The complex biology of aryl hydrocarbon receptor activation in cancer and beyond. *Biochem Pharmacol.* 2023;216:115798. [PubMed: 37696456]
183. Belizaire R, Wong WJ, Robinette ML, Ebert BL. Clonal haematopoiesis and dysregulation of the immune system. *Nat Rev Immunol.* 2023;23(9):595–610. [PubMed: 36941354]
184. Burleson JD, Siniard D, Yadagiri VK, Chen X, Weirauch MT, Ruff BP, et al. TET1 contributes to allergic airway inflammation and regulates interferon and aryl hydrocarbon receptor signaling pathways in bronchial epithelial cells. *Sci Rep.* 2019;9(1):7361. [PubMed: 31089182]

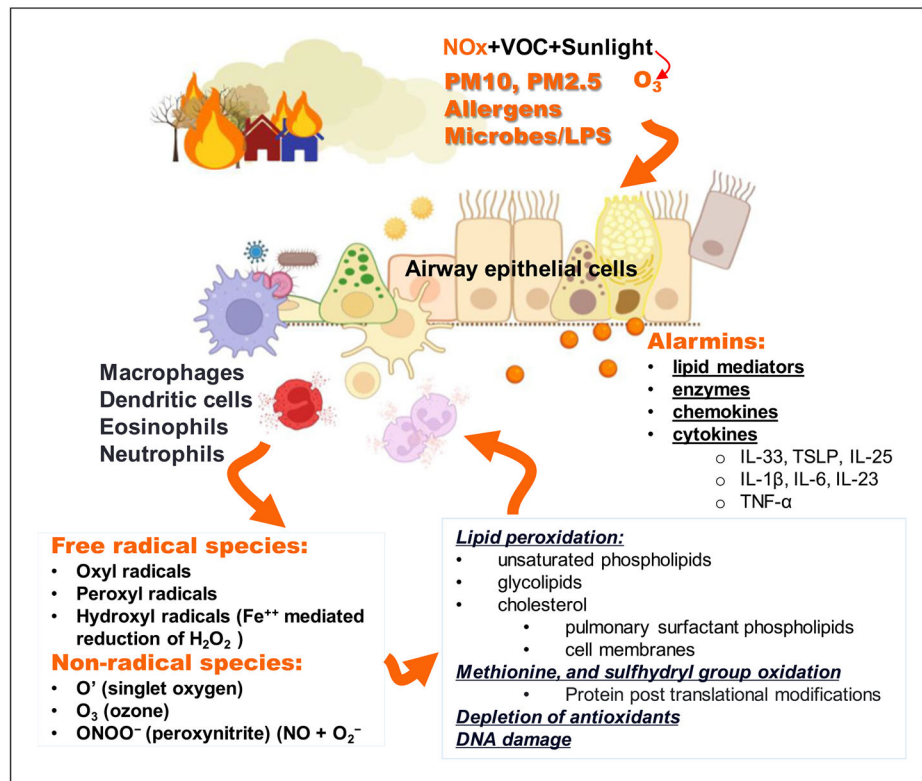
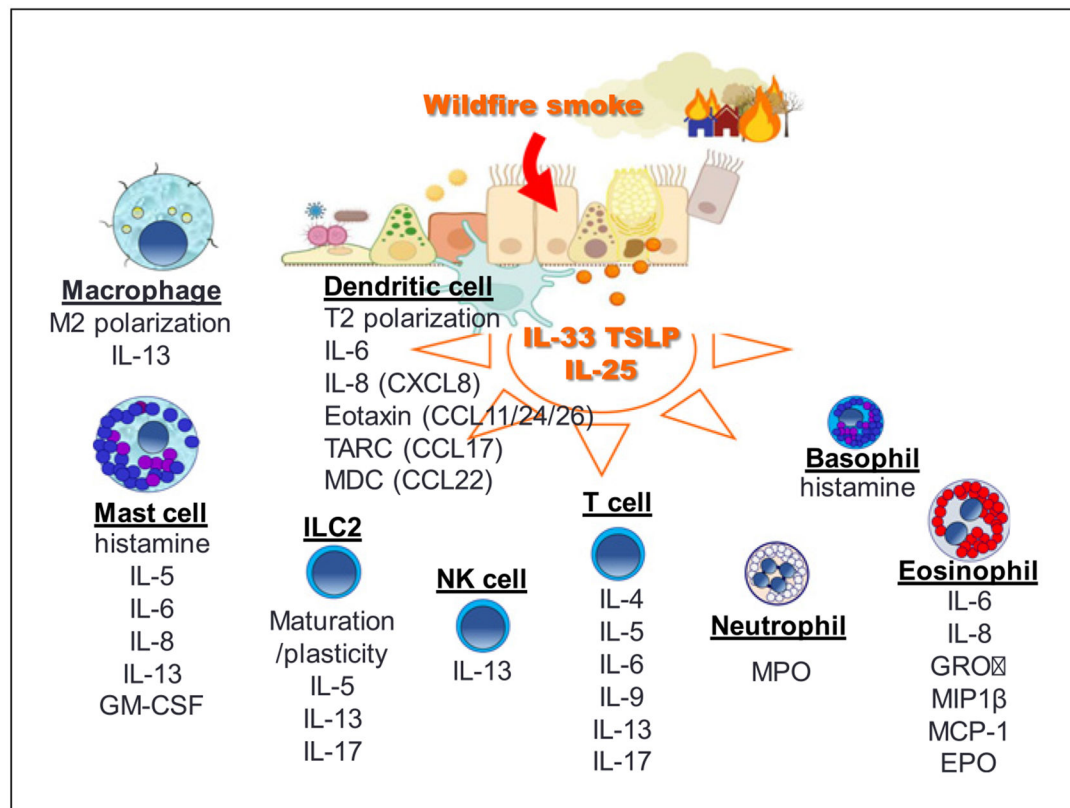


Fig. 1.
 Wildfire-smoke-induced oxidative stress leads to epithelial barrier injury and release of alarmins.

**Fig. 2.**

Wildfire-smoke-induced oxidative stress leads to activation of residential immune and inflammatory cells and to neutrophilic and eosinophilic airway inflammation. This diagram depicts the pluripotent effects of type-2 polarizing alarmins on the immune and inflammatory cells that reside in the airway submucosal tissue.

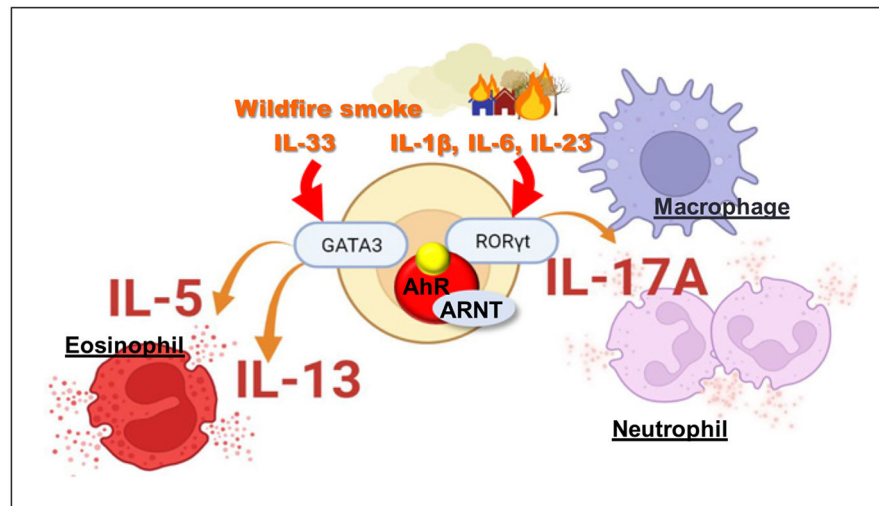
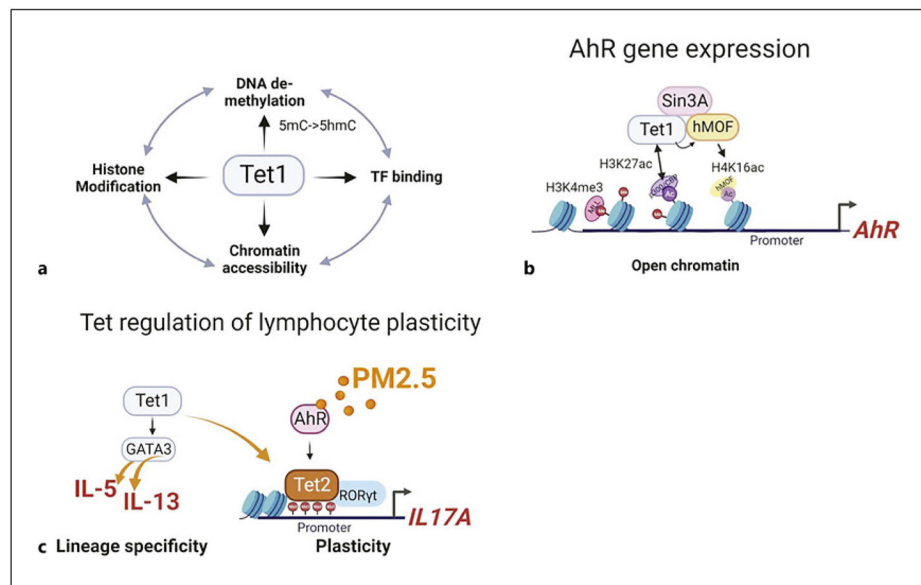


Fig. 3.

Wildfire smoke induces mixed eosinophilic and neutrophilic airway inflammation by stimulating AhR signaling on immune cells. Intracellular AhR in epithelial, myeloid, and lymphoid cells mediates the effects of PAHs and other ligands by transcriptional regulation. Upon ligand binding, the AhR complex translocates into the nucleus and heterodimerizes with AhR nuclear translocator (ARNT) to induce transcription of cytokines such as IL-17A through ROR γ t. The type 2 immune profile (characterized by IL-5 and IL-13 production) in the asthmatic airways may be skewed toward IL-17A production by activating the AhR-ARNT-ROR γ t signaling.

**Fig. 4.**

Tet1 proposed as a potential master epigenetic regulator of genes involved in mediating wildfire smoke effects in asthma. **a** DNA methylation, histone modification, chromatin accessibility, and transcription factor (TF) binding influence each other and coordinately regulate gene expression. It is proposed that Tet proteins regulate all these mechanisms. 5mC, 5-methylcytosine; 5hmC: 5-hydroxymethylcytosine. **b** Tet1 regulates AhR expression through histone modifications and chromatin accessibility. Tet1 is known to form a complex with SIN3A (a scaffold protein) and hMOF (a histone acetyltransferase), which prevents auto-acetylation of hMOF and facilitates it to add H4K16ac and activate gene expression. H3K4me3 (added by MLL) and H3K27ac (added by p300/CBP) are also markers for active transcription. Tet1 can form a complex with p300/CBP. **c** Tet proteins may regulate cytokine production of ILC2 through promoter demethylation, which allows the binding of lineage-specific TFs (e.g., GATA3). In response to PM2.5 exposure, Tet1 may induce expression of IL17A through promoting AhR binding in ILC2 and result in ILC2 plasticity. **d** Tet may regulate promoter methylation of NR3C1 gene and promote the expression of glucocorticoid receptor through demethylation. Reducing the recruitment of Tet proteins to NR3C1 promoter downregulates the expression of the receptor and reduces the responsiveness to steroid.